

1:

```
raghu@Raghu: ~  
GNU nano 8.7.1  
#include <stdio.h>  
#include <unistd.h>  
#include <sys/wait.h>  
  
int main()  
{  
    int fd[2];  
  
    pipe(fd);  
  
    if (fork() == 0)  
    {  
        dup2(fd[1], STDOUT_FILENO);  
  
        close(fd[0]);  
        close(fd[1]);  
  
        execlp("ls", "ls", "-l", NULL);  
    }  
  
    if (fork() == 0)  
    {  
        dup2(fd[0], STDIN_FILENO);  
  
        close(fd[0]);  
        close(fd[1]);  
  
        execlp("grep", "grep", ".c", NULL);  
    }  
  
    close(fd[0]);  
    close(fd[1]);  
  
    wait(NULL);  
    wait(NULL);  
  
    return 0;  
}
```

Output:

```

raghu@Raghu:~$ nano null_pointer.c
raghu@Raghu:~$ nano practical5b.c
raghu@Raghu:~$ gcc practical5b.c -o practical5b
raghu@Raghu:~$ ./practical5b
-rw-r--r-- 1 raghu raghu 699 Sep 2 05:32 FCFS.c
-rw-r--r-- 1 raghu raghu 956 Sep 2 05:35 SJF_NON.c
-rwxr-xr-x 1 raghu raghu 16160 Sep 2 05:53 Sigchild
-rw-r--r-- 1 raghu raghu 617 Sep 2 05:52 Sigchild.c
-rw-r--r-- 1 raghu raghu 552 Sep 2 05:56 Sigpipe.c
-rw-r--r-- 1 raghu raghu 420 Sep 2 05:58 Sigsegv.c
-rw-r--r-- 1 raghu raghu 341 Sep 2 05:51 Sigterm.c
-rw-r--r-- 1 raghu raghu 158 Sep 26 04:57 address.c
-rw-r--r-- 1 raghu raghu 575 Sep 26 05:11 address_translation.c
-rw-r--r-- 1 raghu raghu 1035 Aug 27 03:37 anonymous.c
-rw-r--r-- 1 raghu raghu 220 Aug 27 04:20 fifo.c
-rwxr-xr-x 1 raghu raghu 16048 Aug 27 04:13 fifo_create
-rw-r--r-- 1 raghu raghu 222 Aug 27 04:13 fifo_create.c
-rw----- 1 raghu raghu 39 Aug 27 04:08 fifo_read.c.save
-rw-r--r-- 1 raghu raghu 229 Aug 27 04:02 fifo_write.c
-rw----- 1 raghu raghu 19 Aug 27 03:58 fifo_write.c.save
-rw-r--r-- 1 raghu raghu 532 Aug 6 04:01 hello3.c
-rw-r--r-- 1 raghu raghu 528 Aug 26 05:12 keyboard.c
-rwxr-xr-x 1 raghu raghu 16128 Sep 26 05:14 malloc_array
-rw-r--r-- 1 raghu raghu 382 Sep 26 05:13 malloc_array.c
-rw-r--r-- 1 raghu raghu 463 Sep 26 05:00 memory_segments.c
-rw-r--r-- 1 raghu raghu 168 Sep 26 05:15 null_pointer.c
-rw-r--r-- 1 raghu raghu 424 Sep 26 05:01 page_fault.c
-rwxr-xr-x 1 raghu raghu 16216 Sep 26 05:30 practical5b
-rw-r--r-- 1 raghu raghu 507 Sep 26 05:30 practical5b.c
-rwxr-xr-x 1 raghu raghu 16312 Aug 8 09:57 pratical1
-rw-r--r-- 1 raghu raghu 567 Aug 8 09:57 pratical1.c
-rw-r--r-- 1 raghu raghu 1211 Sep 2 05:37 priority.c
-rw-r--r-- 1 raghu raghu 453 Aug 27 04:21 read.c
-rw-r--r-- 1 raghu raghu 1275 Sep 2 05:39 roundrobin.c
-rw-r--r-- 1 raghu raghu 412 Aug 26 04:56 shell.c
-rw-r--r-- 1 raghu raghu 806 Aug 26 05:17 shellhistory.c
-rw-r--r-- 1 raghu raghu 328 Aug 29 05:18 signal.c
-rwxr-xr-x 1 raghu raghu 16184 Sep 3 03:59 socketclient
-rw-r--r-- 1 raghu raghu 646 Sep 2 02:59 socketclient.c

```

2:

```
raghu@Raghu: ~  
GNU nano 8.7.1  
#include <stdio.h>  
#include <signal.h>  
#include <unistd.h>  
  
void handler(int sig)  
{  
    if (sig == SIGINT)  
        printf("\nSIGINT Received\n");  
  
    if (sig == SIGTERM)  
        printf("\nSIGTERM Received\n");  
  
    if (sig == SIGUSR1)  
        printf("\nSIGUSR1 Received\n");  
}  
  
int main()  
{  
    signal(SIGINT, handler);  
    signal(SIGTERM, handler);  
    signal(SIGUSR1, handler);  
  
    while (1)  
    {  
        printf("Running...\n");  
        sleep(3);  
    }  
  
    return 0;  
}
```

Output:

```
raghu@Raghu:~$ nano practical5b.c  
raghu@Raghu:~$ nano signal_handling.c  
raghu@Raghu:~$ gcc signal_handling.c -o signal_handling  
raghu@Raghu:~$ ./signal_handling  
Running...  
Running...  
Running...  
Running...  
Running...
```